

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 2: from Neural Nets to Data

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Review Lecture 1

- ***Question:** How can we define "artificial intelligence"? What are examples of applications?*

Answer: Methods that enable machines to perceive their environment and use learning and intelligence to take actions to achieve a defined goal. Examples include: recognizing handwritten numbers, generating text, selecting content, ...

- ***Question:** What is "classification"? How does the "pipeline" for supervised learning work?*

Answer: Classification assigns a class to an observation (data point), e.g., one of 10 possible digits. To train a supervised learning algorithm for classification, we need example observations with correct class labels.

- ***Question:** How does a decision tree work?*

Answer: Attributes of observations are nodes of the decision tree, edges are ranges of attribute values, "leaves" are class labels. To classify an observation, we traverse the decision tree from top to bottom, turning right or left at each node depending on the attribute value of the observation.

Overview of Today's Lecture

- 2.1 Neural Networks
- 2.2 Performance of Algorithms
- 2.3 Bias & Variance
- 2.4 Data Types & Data Problems
- 2.5 Regression

3Blue1Brown videos: [What is a neural network?](#), [Gradient Descent](#), [Backpropagation](#).

[A guide to performance metrics in machine learning](#)

Statquest machine learning fundamentals: [Bias and Variance](#)

Machine Learning Mastery: [Regression Metrics](#)
[content for further reading]

Learning Objectives

- Understand how a neural network works.
- Measure the performance of a machine learning algorithm.
- Understand how to avoid underfitting and overfitting during training.
- Have an overview of different data types and potential data problems.